MSDS 451 Assignment 03
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Problem Description

The goal of this assignment is to implement an automated trading solution for a given portfolio. While previous assignments focused on portfolio optimization and time series analysis, this assignment extends these concepts by incorporating automated trading logic. Although trading strategies can achieve high complexity through regime detection and machine learning, this study employs a minimal mean-reversion approach utilizing Bollinger Bands to determine daily buy and sell signals for individual equities.

Data Preparation

Data was ingested from Yahoo Finance for the top 10 holdings of the SPDR S&P Biotech ETF (XBI). These equities were selected due to their relevance to the overarching term project. The selected tickers are MRNA, EXEL, HALO, REGN, AMGN, BIIB, UTHR, GILD, ABBV, and NTRA. The equity portfolio is benchmarked against the SPDR S&P 500 ETF Trust (SPY). Each asset's time series contains daily adjusted Open, High, Low, Close, and Volume (OHLCV) metrics. Rolling means and standard deviations were computed over this dataset to construct the Bollinger Bands.

Research Design

This study evaluates a mean-reversion trading strategy using Bollinger Bands, which operates on the premise that asset prices tend to revert to their historical average over time. Under this strategy, long positions are entered when an asset's price falls below its lower statistical boundary and exited when the price exceeds the upper boundary. These boundaries are established using a 20-day rolling simple moving average (SMA) and a two standard deviation (2σ) envelope.

Backtesting can be executed via two primary paradigms: iterative event-driven simulation (Backtrader) and vectorized calculation (VectorBT). Iterative simulation steps sequentially through time on a daily basis to evaluate strategy rules analogously to a live trading engine; while highly realistic and considered an industry standard, it incurs greater computational overhead. Conversely, vectorized architectures process entire dataset matrices simultaneously, offering substantial computational speedups. Given

the modest portfolio size and timeframe, an event-driven backtesting framework was chosen, though vectorized methods remain advantageous for larger universes spanning thousands of assets.

The annualized Sharpe ratio serves as the primary metric for risk-adjusted performance evaluation. The automated trading strategy is benchmarked against a passive buy-and-hold strategy on SPY across the evaluation period from 2019 to 2026. Trade execution logs generated by the strategy are used to construct an equity curve and compare cumulative performance against the benchmark.

Results



The chart above depicts the daily price trajectory of SPY alongside its associated trading signals over the preceding 252 trading days. The solid line represents the 20-day moving average, the shaded region denotes the ±2σ Bollinger Bands, and the directional markers indicate buy and sell signals. Visualizations for the ten individual portfolio equities are included in the Appendix. These figures illustrate the systematic decision-making process driven by the Bollinger Band parameters. On the SPY benchmark, long positions are generally initiated near local troughs and exited near local peaks. Aside from a notable divergence in April 2026—during which SPY experienced an uncaptured upward movement of approximately $150—the model effectively identified local extrema based on band penetrations.

Strategy	Start Value	End Value	Total Return %	CAGR %	Sharpe (ann.)	Max Drawdown %	Closed Trades	Win Rate %
Bollinger Band (XBI Top 10)	100000.00	157412.35	57.41	5.97	0.66	12.47	335	68.36
Buy & Hold SPY	100000.00	335949.33	235.95	16.76	0.91	33.28	0	-

Ticker	Trades	NetPnL	WinRate	AvgDaysHeld
NTRA	35	11124.91	71.43	11.06
BIIB	38	9578.16	63.16	15.00
MRNA	33	9398.29	69.70	14.73
EXEL	36	8804.32	69.44	12.94
GILD	33	5834.56	75.76	13.00
UTHR	32	4590.07	75.00	15.03
ABBV	31	4285.31	74.19	13.16
AMGN	33	3835.86	66.67	14.85
HALO	33	169.59	54.55	15.36
REGN	31	-307.70	64.52	15.42

Evaluating the Bollinger Band strategy against the passive SPY buy-and-hold baseline highlights the relative performance trade-offs of the approach. The benchmark performed exceptionally well over the evaluation horizon, yielding a total return of 235.95% (16.76% CAGR). However, the benchmark also exhibited a substantially higher maximum drawdown of 33.28%, as a passive buy-and-hold position lacks risk-mitigation mechanics to exit during market downturns. While broad market indices experienced strong secular growth over the seven-year period, the passive strategy's capital appreciation remains significant.

The Bollinger Band strategy generated a cumulative return of 57.41% (5.97% CAGR) while achieving a lower maximum drawdown of 12.47% relative to SPY. Across the portfolio, 335 trades were closed with an aggregate win rate of 68.36%. Given the high volatility of the biotechnology sector, the strategy demonstrated strong trade selectivity, maintaining win rates above 54% across all individual tickers. Nevertheless, high win rates did not uniformly yield positive net profitability, as observed in Regeneron (REGN), which realized a net loss of $307.70 despite a 64.52% win rate. The Net Profit and Loss (NetPnL) metric reflects cumulative net returns after transaction costs.

Additionally, the total trade count and average holding duration remained consistent across assets. For instance, Biogen (BIIB)—which generated the highest trade volume (38 trades)—accumulated an aggregate holding duration of 570 calendar days out of approximately 2,865 total days in the study window. Restricting the evaluation to active trading days (approximately 2,000 days), the portfolio's market exposure rate was approximately 28.5%. Consequently, the portfolio held cash positions for the majority of the timeline, explaining the lower absolute return relative to the fully invested buy-and-hold strategy.

Conclusion

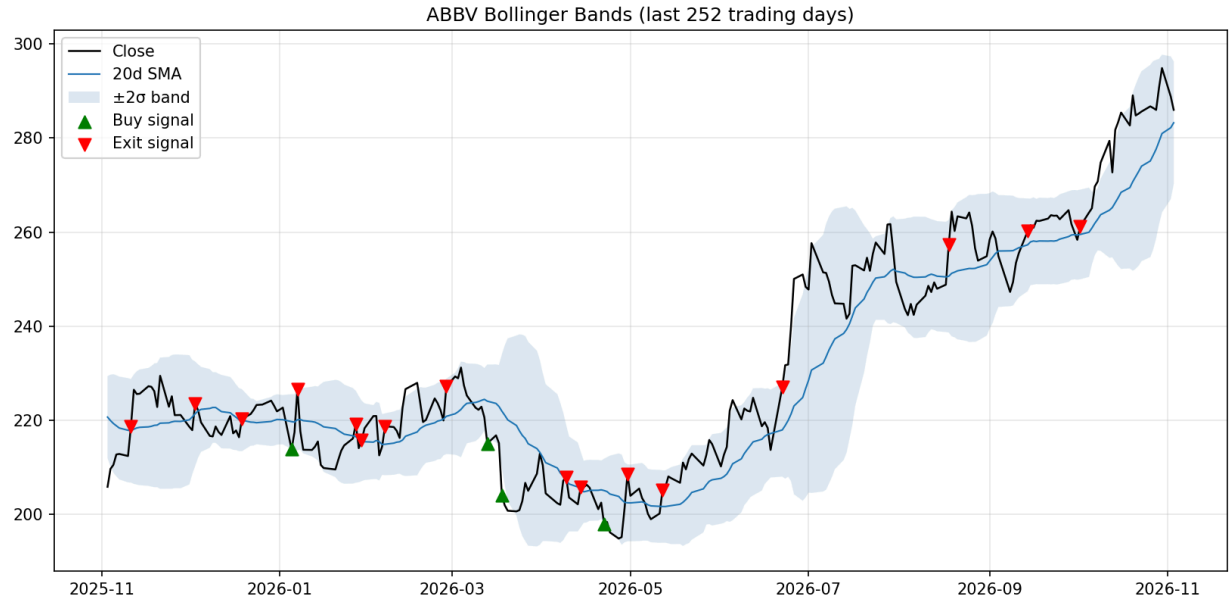
The empirical results demonstrate that a basic mean-reversion framework can capture meaningful market inefficiencies without relying on complex architectures such as Hidden Markov Models or external alternative data streams. The core assumption—that asset prices exhibit mean-reverting behavior around a rolling central tendency—remains surprisingly robust across varying macroeconomic regimes and market shocks. Nevertheless, structural limitations exist. The low trade frequency (averaging 4 to 5 round-trip trades per asset per year) and high cash allocation indicate potential capital inefficiency.

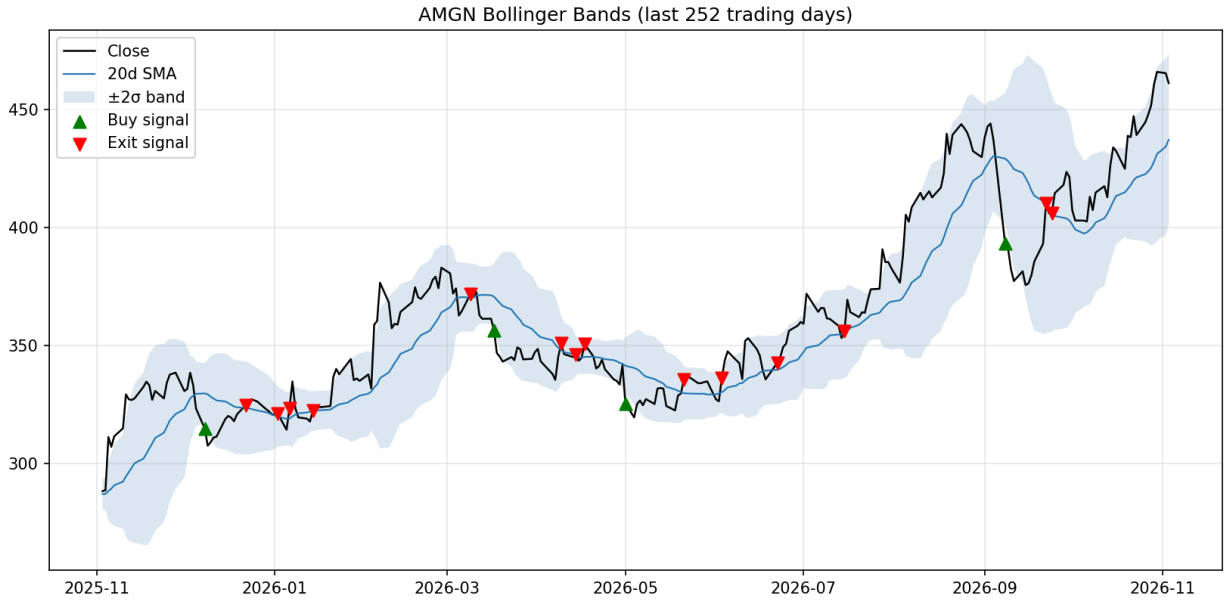
To enhance trade velocity and market engagement, two parameter adjustments are proposed: reducing the rolling lookback window from 20 days to 10 days, and narrowing the bandwidth multiplier k from 2.0 to 1.5. A shorter lookback period increases responsiveness to short-term price dynamics, while narrower bands increase boundary crossing frequency. Additionally, expanding the strategy from a long-only framework to allow short positions could double trade opportunities by monetizing downward mean-reversion trends.

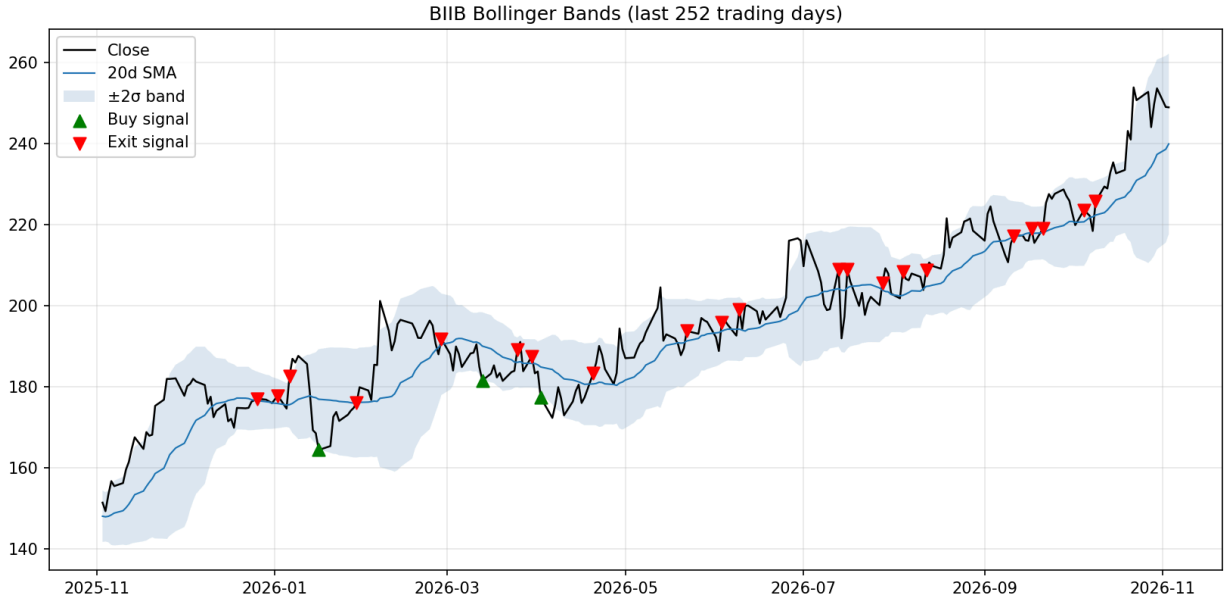
Future work will focus on integrating these automated execution mechanisms with the portfolio optimization models developed in the term project. Incorporating shorting capabilities and multi-asset covariance constraints will provide deeper insights into strategic decision boundaries across portfolio constituents.

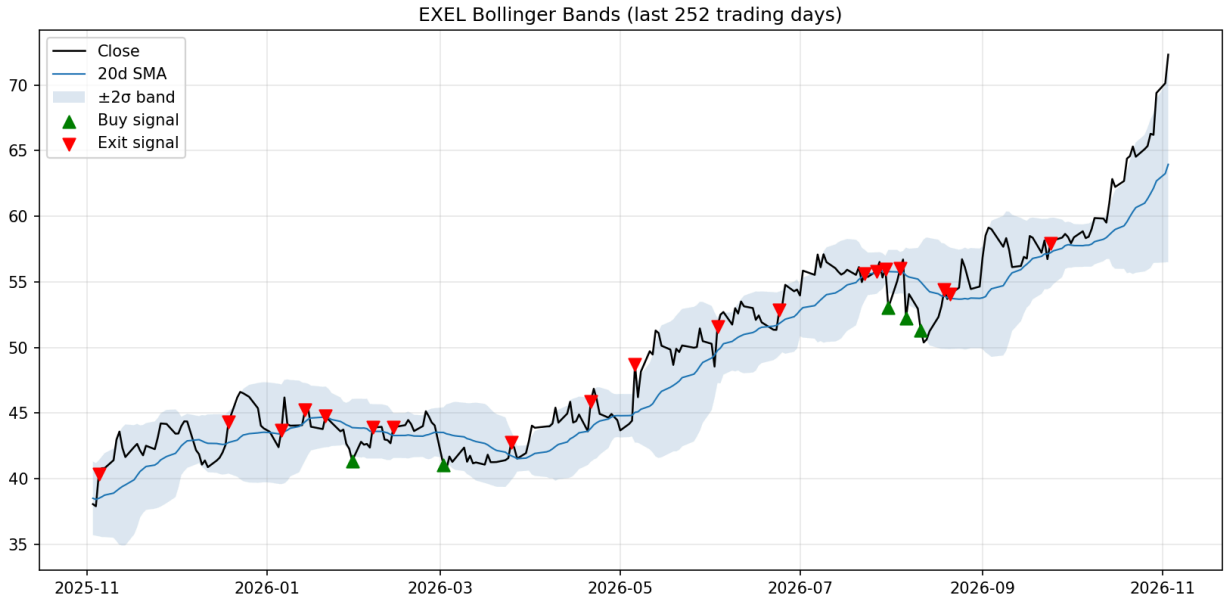
Appendix

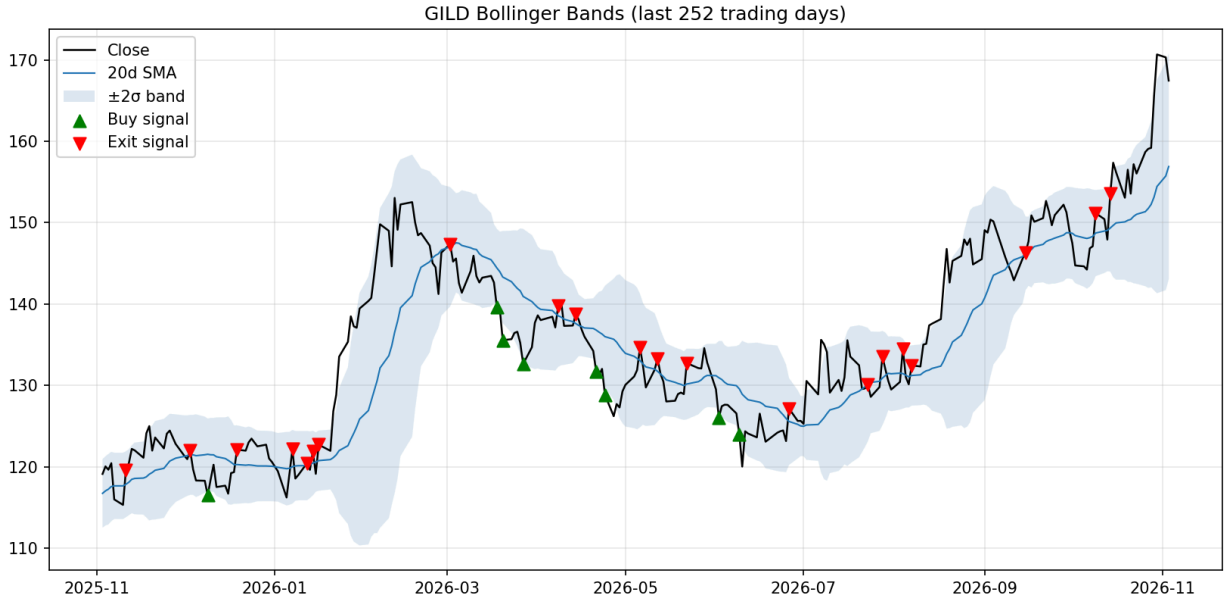
Github: https://github.com/KshaunishSoni123/MSDS451_Assignment03

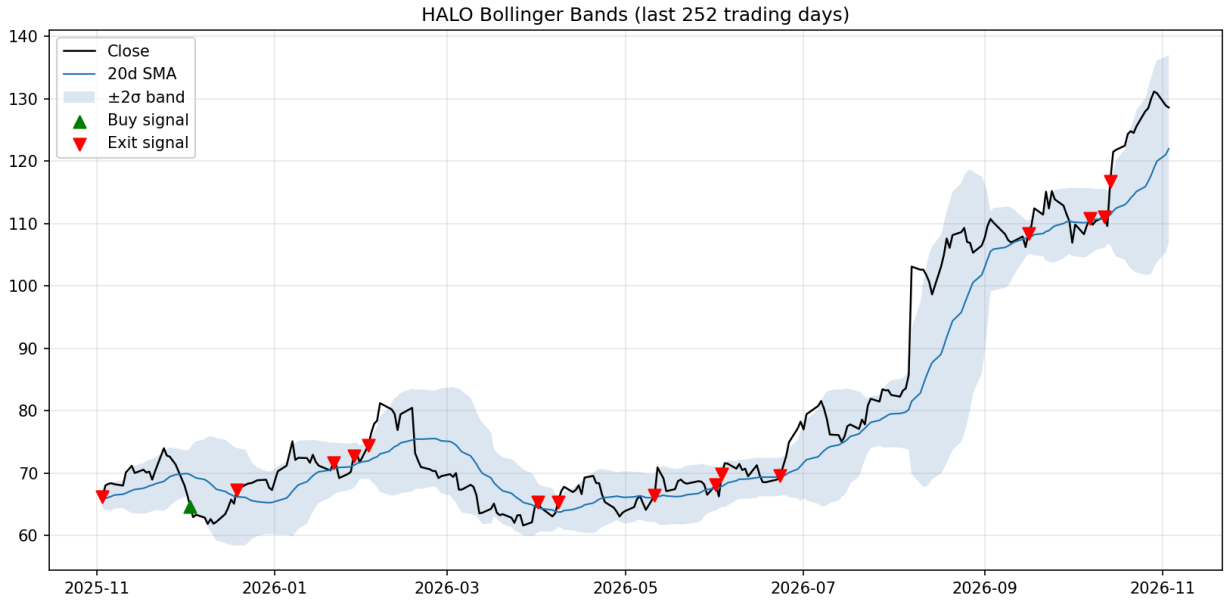


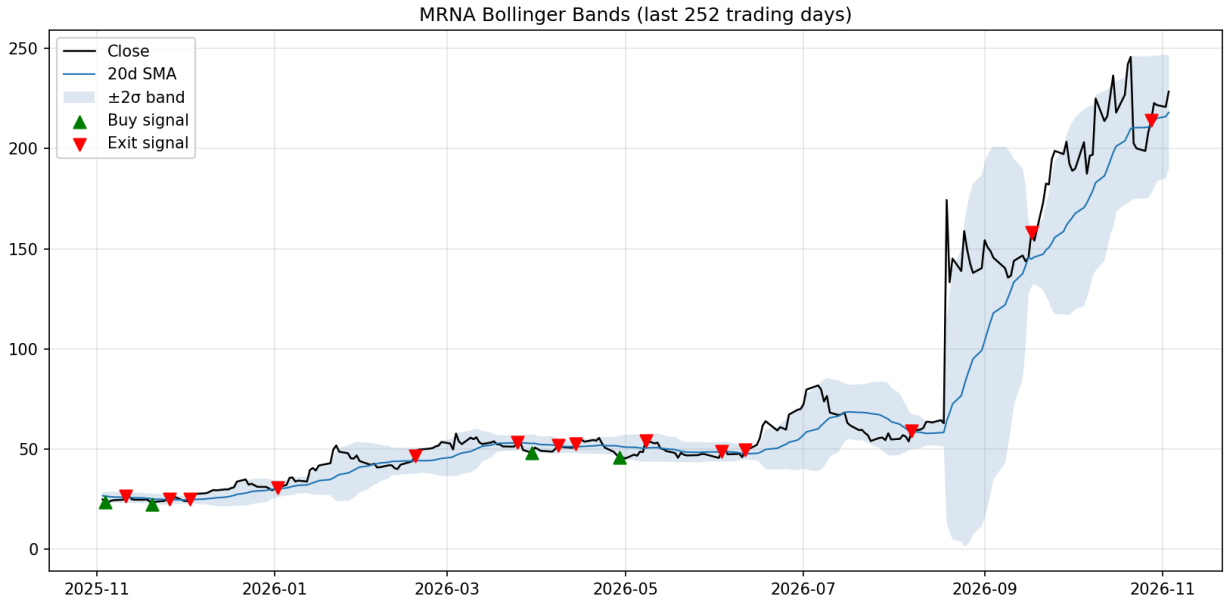

AMGN Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
450
400
350
300
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11

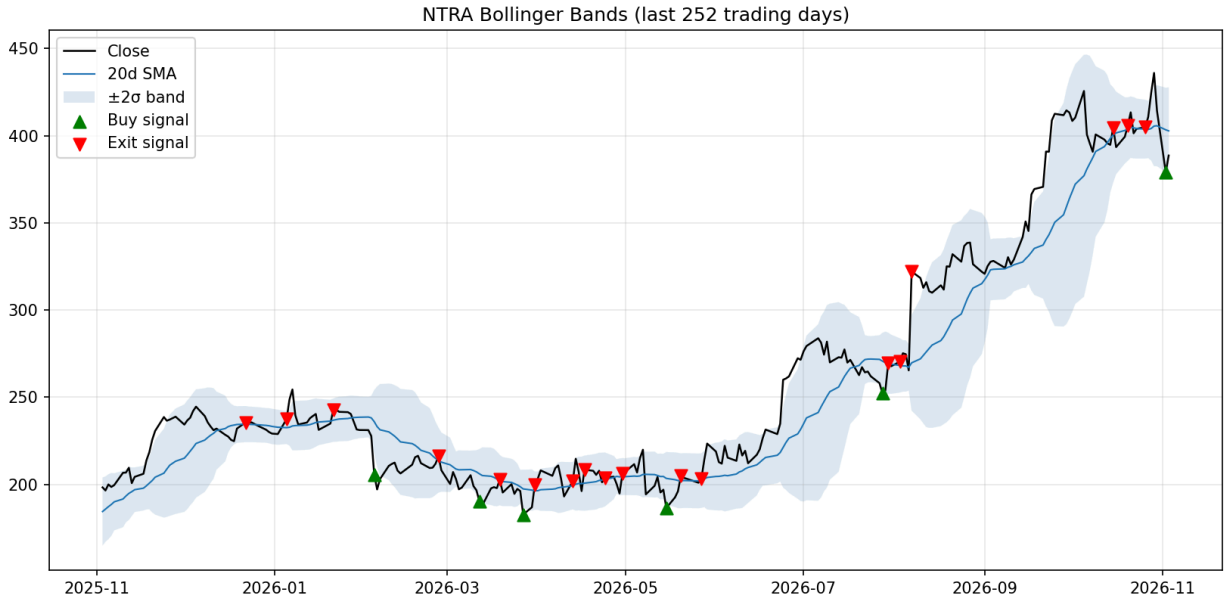

BIIB Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
260
240
220
200
180
160
140
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11

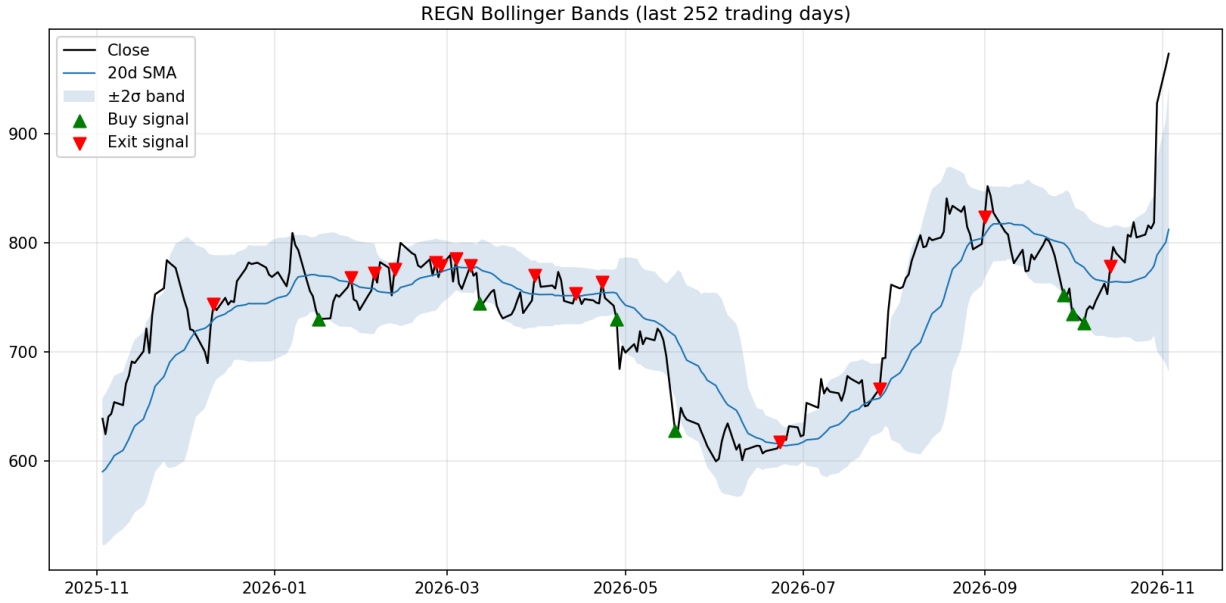

EXEL Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
70
65
60
55
50
45
40
35
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11

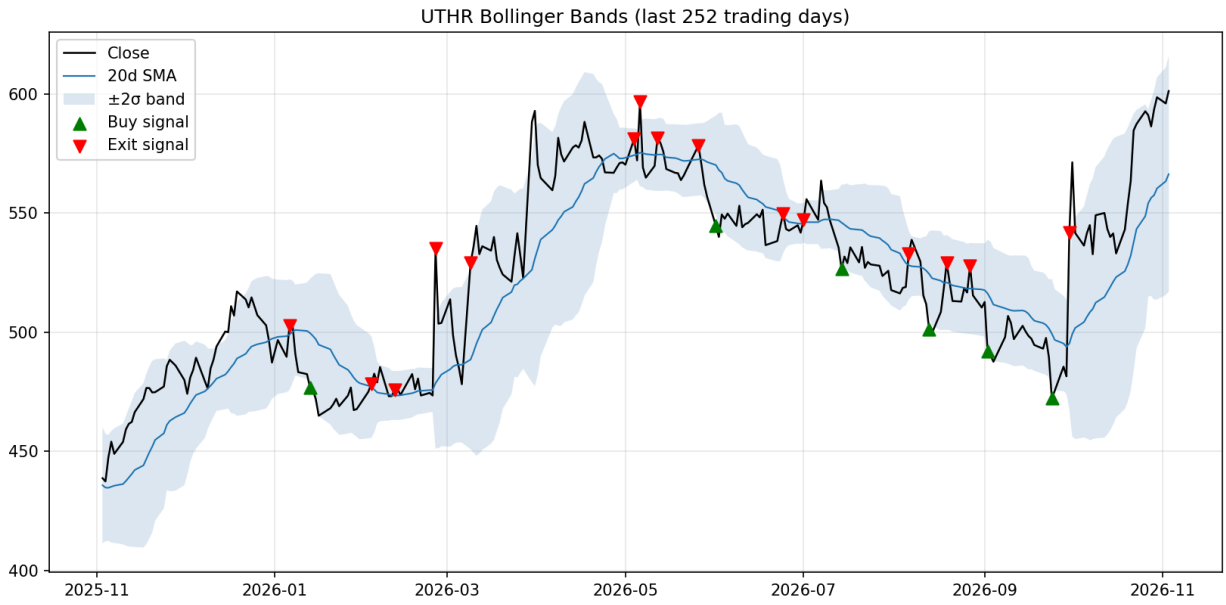

GILD Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
170
160
150
140
130
120
110
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11

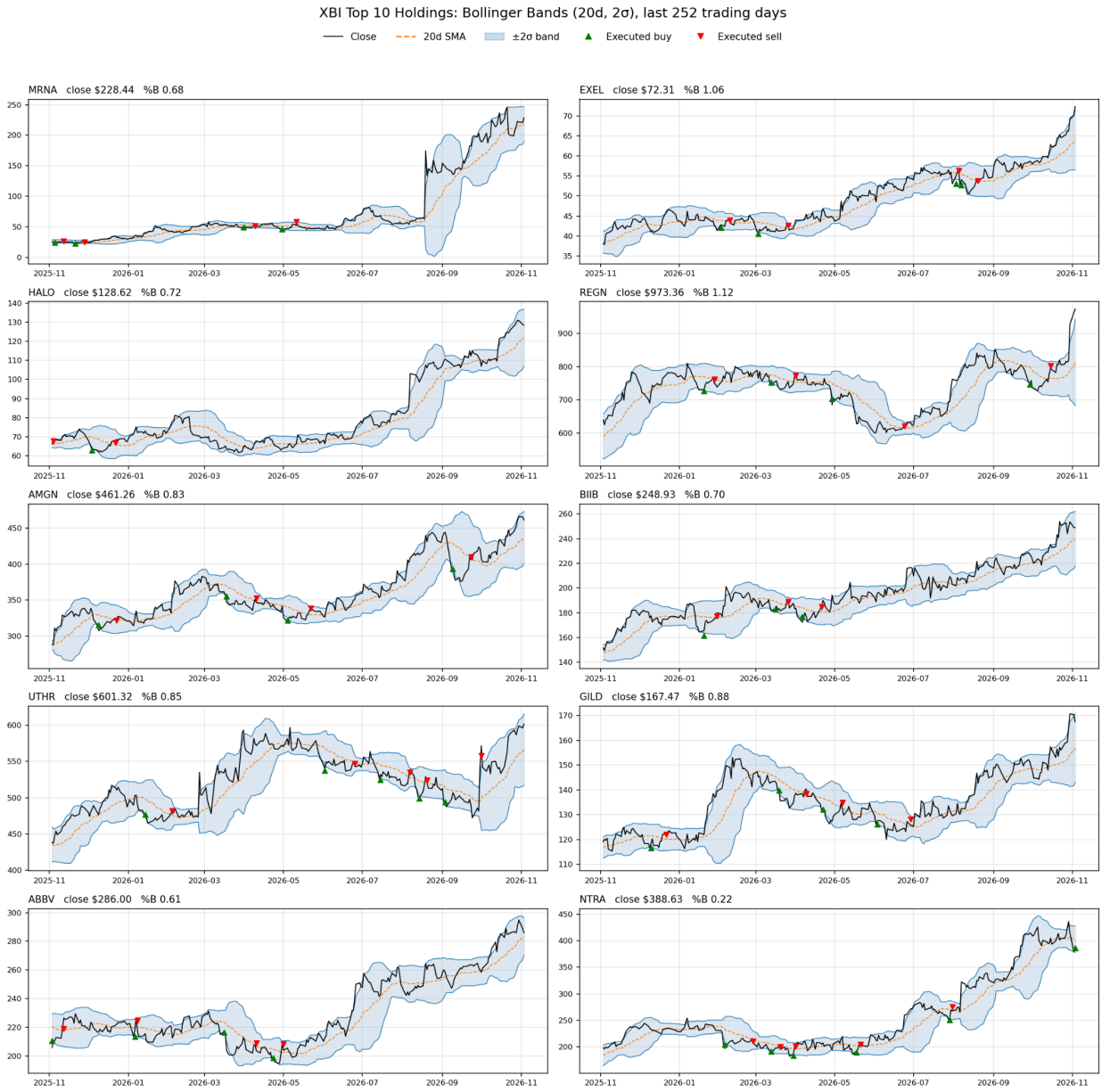

HALO Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
140
130
120
110
100
90
80
70
60
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11


MRNA Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
250
200
150
100
50
0
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11


NTRA Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
450
400
350
300
250
200
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11


REGN Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
900
800
700
600
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11


UTHR Bollinger Bands (last 252 trading days)
Close
20d SMA
±2σ band
Buy signal
Exit signal
600
550
500
450
400
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11


XBI Top 10 Holdings: Bollinger Bands (20d, 2σ), last 252 trading days
Close
20d SMA
±2σ band
Executed buy
Executed sell
MRNA close $228.44 %B 0.68
EXEL close $72.31 %B 1.06
HALO close $128.62 %B 0.72
REGN close $973.36 %B 1.12
AMGN close $461.26 %B 0.83
BIIB close $248.93 %B 0.70
UTHR close $601.32 %B 0.85
GILD close $167.47 %B 0.88
ABBV close $286.00 %B 0.61
NTRA close $388.63 %B 0.22
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09
2026-11